Homework #3, AA2025–2026: Trading and best combination of stocks in portfolio

The stocks are varying in value depending on the single company strategies, success, worldwide economics, etc., it is overall a complex and non-linear dynamic system. One investor might decide to invest in stocks for gaining revenues, and this Homework (Hw) aims to exploit strategies to make the most appropriate decisions for investments using a set of rational rules.

Assume for the Hw that one investor has 10.000 Euros at 1 Jan. 2018 and decides to create a portfolio of 10 stocks with amount $N_1(t)$, $N_2(t)$, ..., $N_{10}(t)$ that are varying in time t, with prices $P_1(t), \ldots, P_{10}(t)$ such that the instantaneous capital on k-th stock is

$$C_k(t) = N_k(t)P_k(t),$$

that depends on the number of stocks $N_k(t)$ that one decides to invest at trading day t.[1] The total capital of the portfolio is the sum of the istantaneous capitals

$$C(t) = \sum_{k=1}^{10} C_k(t) = \sum_{k=1}^{10} N_k(t)P_k(t).$$

At time 1 Jan. 2018 (here considered as $t = 0$) the total capital available for investment is $C(0) = 10.000$ Euro. It is required in this Hw that one invests the overall capital.

To collect stocks data, one can access from WEB (Google search for keywords: *historical stock values*) and download the files from some stocks values vs time over the entire 2018 year from NASDAQ. The portfolio considered herein is mixed but oriented to technology including social stocks (FB, TWTR), software stocks (MSFT, IBM, SNE), technology stocks (QCOM, INTC, AVGO), online market (AMZN), phones (AAPL). From the database there are additional information so that one can extract for the Hw such as the volumes of stocks exchanged on one day $V_1(t), \ldots, V_{10}(t)$ (the amount of stocks exchanged in one specific day t, this data is available at the end of the day): large volumes for one stock compared to the usual trading volumes for the same stock indicates that the specific stock is in high-demand (and thus value $P_k(t)$ increases between opening and closing as many are buying), or stocks are overvalued (and thus value $P_k(t)$ decreases between opening and closing as many are selling the k-th stock).

Stocks values are in daily-basis, and thus herein one can use the following notation: time $t|t$ is the opening time at the beginning of the trading day, and $t+1|t$ is the time at the end of the trading day. The sequence is the following:

$$\cdots \to (t|t) \to (t+1|t) \to (t+1|t+1) \to (t+1|t+2) \to (t+2|t+2) \to \cdots .$$

For every stock the files contains for the day t the opening stock value $P_k(t|t)$, the closing stock value $P_k(t+1|t)$ as result of *intra-day* trading, the volume is available only after closing $V_k(t) = V_k(t+1|t)$. Notice that in general (e.g., due to after-hours trading) the closing value of one stock does not coincide with the opening for next day:

$$P_k(t+1|t+1) \neq P_k(t+1|t).$$

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Goal is to simulate the trading decisions, the actions for every stock and the balance of the entire portfolio with ultimate goal to maximize the capital gain at 31 Dec. 2018 starting from $C(0) = 10^4$ euro. The student might decide to change strategy on trading daily basis that is tipically based on 3 actions: buy, sell stocks or hold (no sell/buy) every day so that, possibly, the capital profit evaluated over the t-th trading day (i.e., value at the closing of the trading day minus value at the opening of the trading day)

$$\Delta C(t) = C(t+1|t) - C(t|t) = \sum_{k=1}^{10} (N_k(t+1|t)P_k(t+1|t) - N_k(t|t)P_k(t|t))$$

is positive and thus the overall capital is increasing. Other metric for the profit is the instantaneous percentage profit

$$g(t) = \frac{C(t+1|t) - C(t|t)}{C(t|t)},$$

[1] Trading day index is consecutive as reflecting the working days Mon–Fri, and it skips holiday gaps.

or the instantaneous percentage profit for the k-th stock

$$g_k(t) = \frac{N_k(t+1|t)P_k(t+1|t) - N_k(t|t)P_k(t|t)}{N_k(t|t)P_k(t|t)}.$$

or other that the student can propose with the final goal of final capital maximization. In order to maximize the capital, and thus the profit $g_k(t)$, it is necessary to decide if buying (increase $N_k(t+1|t)$ starting from the opening stocks $N_k(t|t)$), or selling (decrease $N_k(t+1|t)$ starting from the opening stocks $N_k(t|t)$, and even selling all stocks if losses are expected too large for the t-th trading day to $N_k(t+1|t) = 0$) or even holding the stocks ($N_k(t+1|t) = N_k(t|t)$). Notice that the number of stocks remains the same from closing to opening as it depends on the willingness to sell or buy:

$$N_k(t+1|t+1) = N_k(t+1|t).$$

The decision to buy, sell or hold the stock(s) depends on the profit increasing strategy, to maximize day-by-day the capital profit $\Delta C(t)$ and thus maximize the final gain at 31 Dec. 2018:

$$C(t = 31\text{Dec}2018) - C(t = 01\text{Jan}2018).$$

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Trading strategy decisions can be simulated by using the first semester 01 Jan 2018–30 Jun 2018 for training, and estimating the optimum strategy. Furthermore, since the opening stock value $P_k(t|t)$ is assumed to be known every trading day, the trading decision to be carried out in the Hw is based on the prediction of the action to be carried out to guarantee a profit $g_k(t)$ at the end of the trading day.

1) Estimate $\hat{P}_k(t+1|t)$ based on the available information at the opening $P_k(t|t)$ and the historical (past) data: $P_k(t+1-j|t-j)$, $V(t-j)$ for $j \geq 1$, and **invest all capital on one stock-only** (i.e., all money goes to buy one stock, say $N_k(t) > 0$ and $N_j(t) = 0$ for $\forall j \neq k$). Use a set of linear predictors, one for every stock

$$\hat{P}_k(t+1|t) = b_{k,0} + \sum_{i=1}^{L} b_{k,i} x_{k,i}(t) = \mathbf{b}_k^T \mathbf{x}_k(t),$$

to evaluate at closing time $t+1$ if estimated (predicted) closing value is larger/smaller than opening

$$\hat{P}_k(t+1|t) \gtrless P_k(t|t)$$

and decide to buy ($N_k(t) > N_k(t-1)$) or sell ($N_k(t) < N_k(t-1)$) subject to the profit on total capital $C(t+1)$, and/or the individual profits $g_k(t)$. Trade according to the first 6 months learning stage for training the linear predictor, invest in the most profitable stock (recall, only one) on each time interval and evaluate the final capital $C(t = 31\text{Dec}2018)$. Set of independent entries in $\mathbf{x}_k(t)$ are based on the available information from the past.[2]

2) Use a binary variable $y_k(t) \in \{0, 1\}$ to label the binary condition that is for buying when $P_k(t+1|t) > P_k(t|t)$ and thus $y_k(t) = 1$, or alternatively for selling when $P_k(t+1|t) < P_k(t|t)$ and thus $y_k(t) = 0$. To learn from the training set if for a trading day t it is $\hat{y}_k(t) = 0$ or 1 one might use the logistic regression model as widely used [2,3] that computes the probability for buy or sell:

$$\Pr\{y_k(t) = 1 \mid \mathbf{x}_k(t), \mathbf{b}_k\} = h_{\mathbf{b}_k}(\mathbf{x}_k(t))$$
$$\Pr\{y_k(t) = 0 \mid \mathbf{x}_k(t), \mathbf{b}_k\} = 1 - h_{\mathbf{b}_k}(\mathbf{x}_k(t))$$

from $\mathbf{x}_k(t)$ that has the same meaning as 1), and the sigmoid (or logistic) function is

$$h_{\mathbf{b}_k}(\mathbf{x}_k(t)) = \frac{1}{1 + \exp\left(-\mathbf{b}_k^T \mathbf{x}_k(t)\right)}.$$

a) Evaluate the logistic regression model from the first semester (suggested reading is the "logistic model regression" in folder) used as learning stage.

[2] Predicting variables in $\mathbf{x}_k(t) = [1, x_{k,1}(t), x_{k,2}(t), \ldots, x_{k,L}(t)]^T$ are not necessarily based on past values on the same stock, but perhaps a mixed set of past values, volumes, etc., see ref. [1].

b) Define the best strategy to change and adapt the portfolio mixture, propose a good strategy (recall that $h_{\mathbf{b}_k}(\mathbf{x}_k(t)) \approx 1$ is a definite buy as profitable, $h_{\mathbf{b}_k}(\mathbf{x}_k(t)) \approx 0.5$ is better to hold the stocks and stay as it is, and $h_{\mathbf{b}_k}(\mathbf{x}_k(t)) \approx 0$ is preferable to sell to avoid a loss of profit), perhaps making the volumes $\{N_k(t)\}$ proportional to the relative probabilities from $h_{\mathbf{b}_1}(\mathbf{x}_1(t))$, $h_{\mathbf{b}_2}(\mathbf{x}_2(t))$, ..., $h_{\mathbf{b}_{10}}(\mathbf{x}_{10}(t))$. Train the strategy from the first semester and evaluate the final capital $C(t = 31\text{Dec}2018)$ for different strategies.

Of course, the metric that will be discussed at Hw delivery is the capital increase, and capital fluctuations vs time $C(t)$ from 1 June 2018 till 31 Dec 2018, discussing different strategies. This will give the opportunity to students to better gain insight of economic dynamics of 2018 that have been very hectic ;). By the way, the Hw is purely exemplificative as any buying/selling is affected by a cost for brokers ranging from few Euros to some small percentage on trading.

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Useful references

[1] Özgür İcan and Taha Buğra Çelik, "Stock Market Prediction Performance of Neural Networks: A Literature Review", *International Journal of Economics and Finance*, pp. 100–108, Vol. 9, No. 11, 2017 (available on web).

[2] `https://www.researchgate.net/publication/325995835`$_{Prediction_of_stock_performance_by_using_logistic_regression}$

[3] `https://www.researchgate.net/profile/Shengtao`$_{Sun/publication/224595103_{New_Approach_of_Stock_Price_Trend}}$
$New-Approach-of-Stock-Price-Trend-Prediction-Based-on-Logistic-Regression-Model.pdf$